

Steel Rust Detection with YOLOv8

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1. Problem and scope

This academic prototype identifies candidate visible rust regions on structural steel in inspection photographs. It supports human maintenance review. It cannot measure corrosion depth, section loss or structural capacity. Success for this exercise means a completed 30-epoch baseline, traceable validation results and a reproducible notebook. No deployment accuracy target has been established.

2. Dataset and annotation policy

The source is Steel Corrosion from University of Tebessa on Roboflow Universe [1], adapted as Structural Steel Rust BBoxes, version 1 [2]. The frozen YOLOv8 export contains 440 images and 4,107 stored boxes in one class, rust (ID 0). Export preprocessing applies auto-orientation and stretch resize to 512 x 512, with no export augmentation. Training augmentation settings are recorded in results/train/args.yaml.

The intended policy encloses visible corrosion attached to steel, separating distinct patches where practical. Paint, dirt, shadows and runoff on concrete are excluded. Broad and overlapping inherited boxes remain. The loader removes one duplicate training label.

The notebook converts the original 353/44/43 train/validation/test split into 352 training images (3,400 stored boxes) and 88 validation images (707 boxes). It combines original validation and test images and moves one training image selected with seed 42. The saved manifest records every assignment. There is no independent test set. Exact pixel-duplicate checks passed, but near-duplicate and shared-site leakage remain unaudited.

3. Training and reproducibility

YOLOv8n starts from yolov8n.pt and trains for 30 epochs with batch 16, image size 512 and seed 42. Ultralytics is pinned to 8.3.221. The completed run on 23 September used a Tesla T4, Python 3.13.15 and PyTorch 2.11.0+cu128. The training call took 207.98 seconds. The author confirmed completion from GitHub in a fresh Colab session without dataset credentials, a Drive mount or manual image uploads.

The public GitHub Release [3] supplies the exact dataset ZIP and best.pt, with SHA256 checks in the notebooks. CPU inference also completed on 24 September (run 074023), taking 45.45 seconds after setup with zero training epochs. This reproduces evaluation of saved weights, rather than retraining the model. Minor CPU/GPU numerical differences occur.

4. Validation results

Precision	Recall	mAP50	mAP50-95
35.51%	36.49%	29.23%	12.24%

These standalone validation metrics cover 88 images and 707 annotated instances. Precision/recall use the Ultralytics validation operating point. Prediction examples use confidence 0.25. Low recall and the drop at stricter IoU thresholds indicate substantial misses and localization difficulty. An earlier RF-DETR result used a different split and is not a controlled comparison.

5. Evidence and error analysis

Repository evidence includes five annotation examples, ten validation predictions and predictions on five author photographs taken in public places. The new photographs have no verified ground-truth labels, so they support qualitative review only. One wide canopy image has no detections at confidence 0.25, which does not establish absence of corrosion.

The documented diagnostic review matches saved predictions by descending confidence to unmatched labels at $\text{IoU} \geq 0.50$. It explains three false positives: an overly broad painted-web box (image 14, maximum IoU 0.277), a duplicate box after a stronger prediction already matches the region (image 12), and a small rusty fragment that poorly overlaps a larger inherited label (image 18, maximum IoU 0.082). These are errors against box labels, not necessarily rust-free regions.

Three false negatives illustrate a missed lower edge on red-painted steel (image 10), a thin vertical rust region (image 11) and an upper connection strip (image 16). Color similarity, limited image detail and lighting are plausible contributors, not proven causes. Full filenames, coordinates and overlays are in `docs/error_analysis.md` and `results/evidence/error_analysis/`. The diagnostic matching procedure does not recompute the official metric pipeline.

6. SAM3 exploration

Two Roboflow Smart Select trials used the displayed SAM3 model and Polygon output. On image 6, an initial point selected much of the beam, including apparently intact paint. Additional positive and negative prompts removed much of the central painted area, but imperfect boundaries and disconnected regions remained. On image 327, a single positive point on an isolated brown patch produced a visually close polygon. Screenshots document both outcomes in `docs/sam_exploration.md`.

Clear, isolated patches were easier in these examples. Complex corrosion required human correction. This is a qualitative observation from two cases, without reference masks, measured IoU or timing comparisons. The SAM previews were not incorporated into the frozen release dataset or the reported training run.

7. Prioritized improvements and responsible use

First: audit near-duplicates and group future splits by structure/site. Freeze a new manifest and clearly distinguish results on a changed split from this baseline. **Second:** review a pilot of approximately 30 images to standardize tight rust-patch boundaries and resolve duplicate/component-sized boxes. **Third:** collect approximately 60 rights-cleared examples covering small rust, red paint or difficult lighting, and verified healthy/dirty steel negatives. These are proposed actions, not completed experiments.

The dataset export declares CC BY 4.0. Preserve attribution to the source and adapted project. The author reports taking the five personal photos in public places where photography was permitted. Private client images must be excluded. Dataset attribution, personal-photo reuse terms and software licensing require separate documentation. Human review remains necessary: neither a missed detection nor a positive box can establish structural safety.

Sources and reproducibility records

[1] [University of Tebessa: Steel Corrosion](#). [2] [Adapted dataset, version 1](#). [3] [GitHub Release v1.0](#).
[Project repository](#): full SHA256 values, environment, split manifest, training history, metrics and evidence. Baseline run: 20260923_132304. CPU inference run: 20260924_074023.